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Inventor(s)	FURUSHO; Kentaro

VEHICLE DRIVE SYSTEM

Abstract

A vehicle drive system is provided, which includes a shaft, a motor which rotates the shaft, a resolver which detects a rotation state of the motor, and a shielding member made of a nonmagnetic material which suppresses leakage magnetic flux from the motor discharging to the resolver. The resolver includes a resolver stator extending to a vicinity of a gap formed between a stator coil and a rotor of the motor, and a resolver rotor which rotates with the rotor of the motor. The shielding member includes a first shielding plate disposed in a space between the gap and the resolver rotor so that the first shielding plate extends in the radial direction of the shaft, and a second shielding plate extending in a direction approaching the rotor of the motor from the first shielding plate.

Inventors: **FURUSHO; Kentaro (Aki-gun, JP)**

Applicant: **Mazda Motor Corporation** (Hiroshima, JP)

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a vehicle drive system mounted on a vehicle.

BACKGROUND OF THE DISCLOSURE

[0002] For example, JP5488127B2 discloses a drive system mounted on electric vehicles and hybrid vehicles, which is provided with a driving motor. This driving motor is comprised of a stator fixed to a case, and a rotor disposed rotatable inside the stator. A resolver for detecting a rotational speed of a rotor shaft which supports the rotor is disposed inside the case. The resolver is located inside a coil end of the motor, and includes a resolver stator, a resolver rotor, and a shielding member for suppressing leakage magnetic flux discharged from the motor coil end.

[0003] Since this resolver is disposed inside the motor coil end, the outer diameter of the resolver is smaller than the outer diameter of the rotor of the motor. Thus, the resolver is separated radially inward of a gap formed between the coil and the rotor of the motor.

[0004] Meanwhile, since the leakage magnetic flux from the motor is mainly discharged from this gap, an adverse effect of the leakage magnetic flux may be less, when the outer diameter of the resolver is small as illustrated in JP5488127B2.

[0005] On the other hand, when the stator of the resolver reaches a vicinity of the gap between the stator coil and the rotor of the motor because of the layout inside the drive system, the adverse effect of the leakage magnetic flux on the resolver increases, and therefore, a sufficient measure will be required.

SUMMARY OF THE DISCLOSURE

[0006] The present disclosure is made in view of the problem described above, and one purpose thereof is to keep leakage magnetic flux discharged from a gap between a stator coil and a rotor of a motor from having an adverse effect on a resolver.

[0007] A vehicle drive system includes a shaft, a motor which rotates the shaft, a resolver which detects a rotation state of the motor, and a shielding member made of a nonmagnetic material which suppresses leakage magnetic flux discharging from the motor to the resolver. The resolver includes a resolver stator extending to a vicinity of a gap formed between a stator coil and a rotor of the motor, and a resolver rotor which rotates with the rotor of the motor. The shielding member includes a first shielding plate disposed in a space between the gap and the resolver rotor so that the first shielding plate extends in a radial direction of the shaft, and a second shielding plate extending in a direction approaching the rotor of the motor from the first shielding plate.

[0008] The first shielding plate may be disposed so that a side surface of the first shielding plate opposes the gap. A radially outer edge part of the first shielding plate may reach radially outward of the gap, and a radially inner edge part of the first shielding plate may reach radially inward of the gap.

[0009] The second shielding plate may extend in a direction approaching the rotor of the motor from the radially inner edge part of the first shielding plate with respect to the shaft.

[0010] The second shielding plate may form an annular shape that continues in a circumferential direction of the rotor of the motor.

[0011] The second shielding plate may be disposed radially inward of the gap.

[0012] The vehicle drive may further include a case which accommodates the motor and the resolver. The first shielding plate may be integrally formed with the second shielding plate, and the first shielding plate may be fastened to the case by a common fastening member with the resolver stator.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIG. **1** is a perspective view of a vehicle drive system, when seen from the front.

[0014] FIG. **2** is a front view of the vehicle drive system.

[0015] FIG. **3** is a partial cross-sectional view of the vehicle drive system.

[0016] FIG. **4** is a view schematically illustrating an internal structure of the vehicle drive system.

[0017] FIG. **5** is an enlarged cross-sectional view of a vicinity of an upper part of a stator and a rotor.

[0018] FIG. **6** is an enlarged cross-sectional view of a vicinity of a lower part of the stator and the rotor.

[0019] FIG. **7** is a right side view of a resolver and a shielding member.

[0020] FIG. **8** is a perspective view of the resolver and the shielding member, when seen from the right side.

[0021] FIG. **9** is a cross-sectional view of a fastening part between the resolver and the shielding member.

DETAILED DESCRIPTION OF THE DISCLOSURE

[0022] Hereinafter, one embodiment of the present disclosure is described in detail with reference to the accompanying drawings. Note that the following preferable embodiment is essentially only illustration, and it is not intended to limit the present disclosure, its application, or its use.

[0023] FIGS. **1** and **2** illustrate a vehicle drive system **1** according to one embodiment of the present disclosure. The vehicle drive system **1** is mounted on a vehicle (not illustrated) and drives driving wheels of the vehicle. The vehicle may be an electric vehicle which is propelled only by the power of an electric motor, or may be a hybrid vehicle which is propelled by the power of both an internal combustion engine and the motor. The motor which generates the power for driving the driving wheels is a driving motor. The hybrid vehicle may be a plug-in hybrid vehicle which can be charged by an external electric power source. In this embodiment, the hybrid vehicle is described as the vehicle on which the vehicle drive system **1** is mounted. Therefore, although a driving battery **80** for supplying electric power to a motor **3** and a control unit **90** for controlling the motor **3** are also mounted on the vehicle on which the vehicle drive system **1** is mounted as illustrated in FIG. **3**, since these components are conventionally common knowledge, detailed description thereof is herein omitted. The control unit **90** has an inverter circuit.

[0024] The vehicle drive system **1** is mounted, for example, in an engine room of the vehicle. The engine room may be provided to a front part or a rear part of the vehicle. As illustrated in the drawings, a front-and-rear direction and a left-and-right direction are defined. That is, on the basis of a state where the vehicle drive system **1** is mounted in the engine room of the vehicle, the side which becomes forward of the vehicle is simply referred to as “the front,” the side which becomes rearward of the vehicle is simply referred to as “the rear,” the side which becomes leftward of the vehicle is simply referred to as “the left,” and the side which becomes rightward of the vehicle is simply referred to as “the right.” The left-and-right direction is a vehicle width direction. This definition of direction is only for facilitating the description of this embodiment and does not limit the present disclosure.

[0025] As partially illustrated in FIG. **2** by an imaginary line (two-dot chain line), an engine E is disposed on the right side of the vehicle drive system **1** in the engine room. A crankshaft (not illustrated) of the engine E extends in the left-and-right direction, and the engine E is a so-called “transversely-mounted engine.” The engine E and the vehicle drive system **1** are lined up in the left-and-right direction, the vehicle drive system **1** is disposed on the left side of the engine E, and a left part of the engine E and a right part of the vehicle drive system **1** is coupled by fastening members (not illustrated). The power outputted from the crankshaft of the engine E is inputted into the vehicle drive system **1**. Note that the left-and-right spatial relationship may be reversed, and the lined-up direction of the engine E and the vehicle drive system **1** is not limited in particular. Further, in the case of the electric vehicle, since the engine E is not mounted therein, the above-described engine room is referred to as a "motor room" below.

[0026] As illustrated in FIG. **3**, the vehicle drive system **1** is provided with a shaft **2**, the motor **3** which rotates the shaft **2**, a resolver **4** which detects a rotation state of the motor **3**, a shielding member **5** which suppresses leakage magnetic flux discharging from the motor **3** to the resolver **4**, and a case A. The shaft **2**, the motor **3**, the resolver **4**, and the shielding member **5** are disposed inside the case A, and therefore, the case A is a member which accommodates the shaft **2**, the motor **3**, the resolver **4**, and the shielding member **5**. The shaft **2** extends in the left-and-right direction inside the case A, and therefore, the axial direction of the crankshaft of the engine E is in agreement with the axial direction of the shaft **2** (illustrated by a reference character "B").

[0027] The case A is provided with a motor housing **10**, a right cover member **20**, and a left cover member **30**. The shaft **2** is rotatably supported by the case A. The motor housing **10**, the right cover member **20**, and the left cover member **30** are made of light alloy, such as aluminum alloy, and are high-rigidity members. In this embodiment, since the right cover member **20** and the left cover member **30** are disposed on the right side and the left side, respectively, they are referred to as such names. However, the right cover member **20** may be referred to as a "front cover member," and the left cover member **30** may be referred to as a "rear cover member." The right cover member **20** is a first cover member, and the left cover member **30** is a second cover member.

[0028] The motor housing **10**, the right cover member **20**, and the left cover member **30** constitute one case A. Since the right cover member **20** is located on the right side of the case A and this right cover member **20** constitutes the right part of the case A, the right cover member **20** may also be referred to as a "right constituent member." Similarly, since the left cover member **30** is located on the left side of the case A and this left cover member **30** constitutes the left part of the case A, the left cover member **30** may also be referred to as a "left constituent member." Since the motor housing **10** is located between the right cover member **20** and the left cover member **30**, and this motor housing **10** constitutes an intermediate part of the case A in the left-and-right-direction, the motor housing **10** may also be referred to as an "intermediate part constituent member." The motor housing **10** is sandwiched between the right cover member **20** and the left cover member **30** in the left-and-right direction. The structure of the case A may further be provided with another member, without being limited to the structure described above.

[0029] As illustrated by the outline structure in FIG. **3**, a first coupling mechanism **2***a* coupled to the crankshaft of the engine E is fixed to a right end part of the shaft **2**. The first coupling mechanism **2***a* is changeable between a connected state in which the crankshaft of the engine E is connected to the shaft **2** and a non-connected state in which the crankshaft of the engine E is not connected to the shaft **2**, and the change operation is controlled by the control unit **90**.

[0030] A second coupling mechanism **2***b* is fixed to the shaft **2**, on the left side of the first coupling mechanism **2***a*. The second coupling mechanism **2***b* is changeable between a connected state in which the motor **3** is connected to the shaft **2** and a non-connected state in which the motor **3** is not connected to the shaft **2**, and the change operation is controlled by the control unit **90**. By changing the second coupling mechanism **2***b* into the connected state, the power of the motor **3** is transmitted to the shaft **2**. By the control unit **90** controlling the first coupling mechanism **2***a* and the second coupling mechanism **2***b*, only the power of the engine E can be outputted, only the power of the motor **3** can be outputted, or the synthesized power of the engine E and the motor **3** can be outputted. For this control technique, conventional well-known techniques can be used. Further, a transmission **2***c* is provided to the shaft **2**, on the left side surface of the second coupling mechanism **2***b*. The power of the engine E and/or the power of the motor **3** are inputted into the transmission **2***c*.

[0031] FIG. **4** is a view schematically illustrating an inner structure of the vehicle drive system **1**, where the first coupling mechanism **2***a* and the second coupling mechanism **2***b* are omitted. A first gear **100** is provided to an intermediate part of the shaft **2**. The output of the transmission **2***c* is transmitted to the first gear **100**. A lower shaft **101** is provided below the shaft **2**, and this lower shaft **101** is also disposed inside the case A. The lower shaft **101** and the shaft **2** are parallel to each other. Both end parts of the lower shaft **101** are rotatably supported inside the case A. A second gear **102** which meshes with the first gear **100** is provided to an intermediate part of the lower shaft **101**.

[0032] A final gear **103** is provided below the shaft **2**. The final gear **103** is fixed to a gear carrier **104**, and this gear carrier **104** is rotatably supported inside the case A. The axial center of the gear carrier **104** is parallel to the axial center of the lower shaft **101**. A third gear **105** which meshes with the final gear **103** is provided to the intermediate part of the lower shaft **101**.

[0033] Therefore, when the power of the engine E and/or the power of the motor **3** is outputted from the transmission **2***c*, it is then transmitted to the second gear **102** via the first gear **100**. Since the second gear **102** and the third gear **105** are fixed to the lower shaft **101**, the rotational force transmitted to the second gear **102** is transmitted from the third gear **105** to the final gear **103** to rotate the gear carrier **104**. Since left and right drive shafts (not illustrated) are coupled to the gear carrier **104** via a differential mechanism (not illustrated), the power of the engine E and/or the power of the motor **3** are transmitted from the drive shafts to the driving wheels (not illustrated).

[0034] As partially illustrated in FIG. **3**, a right end part (axial one end part) of the lower shaft **101** is rotatably supported by a bearing part **200**. Although not illustrated, a left end part (axial other end part) of the lower shaft **101** is rotatably supported by another bearing. The bearing part **200** has a bearing **201** and a bearing fixing part **202** for fixing the bearing **201** to a given position. The bearing fixing part **202** may be formed, for example, integrally with the case A, or it may be formed by a different member from the case A and may be assembled to the case A. The bearing fixing part **202** of this embodiment has a concaved shape which opens leftwardly, and the bearing **201** is inserted into this open part and is assemble to the bearing fixing part **202**.

[0035] The right side of the bearing fixing part **202** is formed so that it axially approaches to the motor **3**. In detail, in order to form the concaved shape into which the bearing **201** is insertable, the bearing fixing part **202** has a bulged surface **202***a* having a shape bulged on the right side. The bulged surface **202***a* and the side surface of the motor **3** have a spatial relationship in which they oppose to each other.

[0036] The fundamental structure of the motor **3** may be common structures which have been conventionally well-known. That is, the motor **3** has a stator coil **3***a* and a rotor **3***b*. The stator coil **3***a* is structured so that a plurality of coils are disposed annually, and is fixed to an inner surface of the case A. On the other hand, the rotor **3***b* includes a plurality of magnets, and is fixed to an outer circumferential surface of the second coupling mechanism **2***b* in a state where it is disposed radially inward of the stator coil **3***a*.

[0037] FIG. **5** illustrates an enlarged view of the vicinity of an upper part of the stator coil **3***a* and the rotor **3***b*, and FIG. **6** illustrates an enlarged view of the vicinity of a lower part of the stator coil **3***a* and the rotor **3***b*. As illustrated in FIGS. **5** and **6**, since an outer circumferential surface of the stator coil **3***a* is separated from an inner circumference surface of the rotor **3***b* in the radial direction, a given gap C is formed therebetween. As illustrated in FIG. **6**, the bulged surface **202***a* of the bearing part **200** is located so that it opposes to the left surface of the stator coil **3***a* and the left surface of the rotor **3***b*, and it also opposes to the gap C formed between the stator coil **3***a* and the rotor **3***b*. Further, in order to shorten the dimension of the case A in the left-and-right direction as much as possible, the interval of the bearing part **200**, and the stator coil **3***a* and the rotor **3***b* is set narrowly.

[0038] The resolver **4** has a resolver stator **40** and a resolver rotor **41**. As illustrated in FIGS. **7** and **8**, the resolver stator **40** has a body part **40***a* comprised of an annular plate member, and a sensor part **40***b* fixed to the body part **40***a*. As illustrated in FIGS. **5** and **6**, the body part **40***a* is fixed to the case A and extends in the radial direction of the shaft **2**. The radially outward part of the body part **40***a* extends to a vicinity of (near) the gap C formed between the stator coil **3***a* and the rotor **3***b* of the motor **3**.

[0039] The resolver rotor **41** is comprised of an annular plate member disposed radially inward of the resolver stator **40**. The radially inward part of the resolver rotor **41** is fixed to the left side surface of the second coupling mechanism **2***b*. Therefore, the resolver rotor **41** is integrally formed with the rotor **3***b* of the motor **3** via the second coupling mechanism **2***b*, and rotates with the rotor **3***b*.

[0040] A concavo-convex shape is formed continuously in the circumferential direction in the radially outward part of the resolver rotor **41**. When the resolver rotor **41** excited by alternating current is rotated with the rotor **3***b* of the motor **3**, the sensor part **40***b* of the resolver stator **40** detects a change in the electromotive force. Based on the concavo-convex shape of the resolver rotor **41** and a relative rotating angle change of the sensor part **40***b*, the rotation position of the rotor **3***b* of the motor **3** can be detected. This detection technique of the resolver **4** may be a conventionally well-known technique.

[0041] During rotation of the motor **3**, the leakage magnetic fluxes (illustrated by arrows D in FIG. **5**) are discharged from the gap C formed between the stator coil **3***a* and the rotor **3***b*. In this embodiment, since the dimension of the case A in the left-and-right direction is desirable to be shortened as much as possible, the distance between the resolver **4**, and the stator coil **3***a* and the rotor **3***b* is set short. Further, the dimension of the case A in the left-and-right direction is also shortened by disposing the second coupling mechanism **2***b* radially inward of the resolver **4**. Thus, the premise structure of this embodiment is a structure where the gap C formed between the stator coil **3***a* and the rotor **3***b* opposes to the resolver **4** with an approached distance therebetween, and therefore, the leakage magnetic flux discharged from the gap C tends to give an adverse effect on the resolver **4**. The leakage magnetic flux may be overlapped with the detection signal as noise, when the sensor part **40***b* of the resolver stator **40** detects the change in the electromotive force.

[0042] Regarding this, in this embodiment, the shielding member **5** is provided so that the leakage magnetic flux discharged from the gap C formed between the stator coil **3***a* and the rotor **3***b* does not to give the adverse effect on the resolver **4**. The shielding member **5** is made of nonmagnetic material, and it is constructed to cover the resolver **4**. Although the nonmagnetic material which constitutes the shielding member **5** is not limited in particular, it may be aluminum alloy, for example.

[0043] The shielding member **5** has a first shielding plate **51** which is disposed so that it extends in the radial direction of the shaft **2** in a space between the gap C formed between the rotor **3***b* the stator coil **3***a*, and the resolver rotor **41**, and a second shielding plate **52** which extends in a direction approaching the rotor **3***b* of the motor **3** from the first shielding plate **51**. The first shielding plate **51** has an annular shape. The right side surface of the first shielding plate **51** is disposed so that it opposes to the gap C formed between the stator coil **3***a* and the rotor **3***b*. The radially outer edge part of the first shielding plate **51** reaches radially outward of the gap C, and the radially inner edge part of the first shielding plate **51** reaches radially inward of the gap C. Therefore, when seen from the left side, it has the spatial relationship in which the gap C is located at an intermediate part of the first shielding plate **51** in the radial direction. The thickness of the first shielding plate **51** is set thinner than the thickness of the body part **40***a* of the resolver stator **40**.

[0044] The second shielding plate **52** extends in a direction approaching the rotor **3***b* of the motor **3** from the radially inner edge part (with respect to the shaft **2**) of the first shielding plate **51**, and forms an annular shape which continues in the circumferential direction of the rotor **3***b* of the motor **3**. An angle formed between the first shielding plate **51** and the second shielding plate **52** is set as about 90°. Note that the angle formed between the first shielding plate **51** and the second shielding plate **52** is not limited to 90°, and it may be set within a range of 80° or more and 120° or less.

[0045] The second shielding plate **52** is disposed radially inward of the gap C formed between the stator coil **3***a* and the rotor **3***b*. Therefore, the second shielding plate **52** can suppress leakage magnetic flux from being discharged from the gap C toward the body part **40***a* of the resolver stator **40** reaches the body part **40***a*.
[0046] The first shielding plate **51** and the second shielding plate **52** are formed integrally. Although the method of integrally forming the shielding member **5** is not limited in particular, the shielding member **5** having the first shielding plate **51** and the second shielding plate **52** can be obtained, for example, by press-molding a sheet of metal plate with a metallic mold (not illustrated). Note that the first shielding plate **51** and the second shielding plate **52** may be constituted from different members. In this case, after forming the first shielding plate **51** and the second shielding plate **52** from the different members, the members may be integrally coupled to each other.
[0047] As illustrated in FIG. **9**, the first shielding plate **51** of the shielding member **5** is fastened to the case A by fastening members **400** common to the resolver stator **40**. That is, a first insertion hole **51***a* into which the shank of a bolt or a screw as the fastening member **400** is inserted is formed in the first shielding plate **51** so that it penetrates in the thickness direction. Further, a second insertion hole **40**A is formed in the body part **40***a* of the resolver stator **40** so that it penetrates in the thickness direction and is in agreement with the first insertion hole **51***a*. A threaded hole A**1** corresponding to the first insertion hole **51***a* and the second insertion hole **40**A is formed in the case A. After inserting the shank of the fastening member **400** into the first shielding plate **51**, the first insertion hole **51***a*, and the second insertion hole **40**A of the resolver stator **40** in this order, the shank is threadedly engaged with the threaded hole A**1** of the case A to fasten the first shielding plate **51** and the resolver stator **40** to the case A together. Although in this embodiment the number of fastening members **400** are four, and the first shielding plate **51** is fastened by the four fastening members **400** to the case A at a plurality of locations which are away in the circumferential direction of the shaft **2** as illustrated in FIGS. **7** and **8**, the number of fastening members **400** is not limited to four, and it may be an arbitrary number, such as three or less or five or more. The same numbers of first insertion holes **51***a*, second insertion holes **40**A, and threaded holes A**1** as the number of fastening members **400** may be formed. Note that the first shielding plate **51** and the resolver stator **40** may be fastened to the case A by different fastening members (not illustrated), without being fastened together.
[0048] As described above, since in this embodiment the interval between the bearing part **200**, and the stator coil **3***a* and the rotor **3***b* is narrow, it is difficult to dispose the first shielding plate **51** of the shielding member **5** between the bearing part **200**, and the stator coil **3***a* and the rotor **3***b*. A notch **51***c* for avoiding interference with the bearing part **200** is formed radially outward of the first shielding plate **51** of the shielding member **5**. Since the bearing part **200** is provided below the shaft **2** and the bearing fixing part **202** is close to the stator coil **3***a* and the rotor **3***b*, the notch **51***c* is formed only in a lower part of the first shielding plate **51** to avoid interference with the bearing fixing part **202**.
[0049] Thus, the notch **51***c* is formed only in the first shielding plate **51**, and is not formed in the second shielding plate **52**. That is, since the notch **51***c* is formed in a part of the shielding member **5** other than the second shielding plate **52**, the second shielding plate **52** can form the annular shape which continues in the circumferential direction. Therefore, the second shielding plate **52** can improve the shielding effects on the leakage magnetic flux. Although the shape of the notch **51***c* may be set arbitrarily, it is the annular shape in this embodiment.
[0050] Further, the notch **51***c* is formed in the part separated from the parts for the fastening by the fastening members **400** in the circumferential direction of the shaft **2**. In other words, the notch **51***c* is formed between the fastening parts by the fastening members **400**. Since the notch **51***c* is the part for avoiding interference with the bearing part **200**, it is preferred to be a notch within the minimum area which can avoid the interference with the bearing part **200**, but it may be formed in a

sufficiently larger area than the bearing part **200**, without being limited to the configuration described above.

[0051] Although it is not illustrated, a gear may be provided at a part in the case A where it interferes with the shielding member **5**. In this case, a notch may be formed for avoiding interference with the gear in the shielding member **5**. Alternatively, both the notch **51***c* for avoiding interference with the bearing part **200** and the notch for avoiding interference with the gear may be formed in the shielding member **5**.

[0052] As illustrated in FIGS. **6** and **8**, a notch **40***c* is also formed in the body part **40***a* of the resolver stator **40** to correspond to the notch **51***c* of the shielding member **5**. Note that the notch **40***c* of the resolver stator **40** is not essential, but it may be provided as needed.

Operation and Effects of Embodiment

[0053] As described above, according to this embodiment, the shielding member **5** is provided inside the case A, the first shielding plate **51** is disposed in the space between the gap C formed between the stator coil **3***a* and the rotor **3***b* of the motor **3**, and the resolver rotor **41**, and the second shielding plate **52** extending in the direction approaching the side surface of the rotor **3***b* of the motor **3** from the first shielding plate **51** is formed. Therefore, the first shielding plate **51** can suppress that the leakage magnetic flux discharged from the gap C formed between the stator coil **3***a* and the rotor **3***b* of the motor **3** reaches the resolver **4**, and the second shielding plate **52** can also suppress the same. Further, since the first shielding plate **51** extends in the radial direction and the second shielding plate **52** extends in the direction approaching the rotor **3***b* of the motor **3**, and the shielding plates **51** and **52** extend in different directions, the shielding effects on the leakage magnetic flux discharged from the gap C increase.

[0054] Further, since the notch **51***c* is formed in the shielding member **5**, interference between the shielding member **5** and the bearing part **200** can be avoided. Thus, when narrowing the interval between the bearing part **200** and the shielding member **5** which are disposed inside the case A in order to shorten the size of the case A in the left-and-right direction, the shielding member **5** can be disposed while avoiding the interference with the bearing part **200**. Further, since the notch **51***c* is formed partially in the shielding member **5**, the shielding effects on the leakage magnetic flux by the shielding member **5** hardly fall, even if the notch **51***c* is provided.

[0055] The above embodiment is merely illustration, and it must not be interpreted restrictively in any aspects. Further, any modifications and changes belonging to the equivalent range of the appended claims fall within the scope of the present disclosure.

INDUSTRIAL APPLICABILITY

[0056] As described above, the vehicle drive system according to the present disclosure can be utilized, for example, for hybrid vehicles or electric vehicles.

[0057] It should be understood that the embodiments herein are illustrative and not restrictive, since the scope of the invention is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof, are therefore intended to be embraced by the claims.

DESCRIPTION OF REFERENCE CHARACTERS

[0058] **1** Vehicle Drive System [0059] **2** Shaft [0060] **3** Motor [0061] **4** Resolver [0062] **40** Resolver Stator [0063] **41** Resolver Rotor [0064] **5** Shielding Member [0065] **51** First Shielding Plate [0066] **51***c* Notch [0067] **52** Second Shielding Plate [0068] **100** First Gear [0069] **101** Lower Shaft [0070] **102** Second Gear [0071] **200** Bearing Part [0072] **201** Bearing [0073] **202** Bearing Fixing Part [0074] A Case [0075] C Gap [0076] **400** Fastening Member

Claims

1. A vehicle drive system, comprising: a shaft; a motor configured to rotate the shaft; a resolver configured to detect a rotation state of the motor; and a shielding member made of a nonmagnetic material and configured to suppress leakage magnetic flux discharging from the motor to the resolver, wherein the resolver includes a resolver stator extending to a vicinity of a gap formed between a stator coil and a rotor of the motor, and a resolver rotor configured to rotate with the rotor of the motor, and wherein the shielding member includes a first shielding plate disposed in a space between the gap and the resolver rotor so that the first shielding plate extends in a radial direction of the shaft, and a second shielding plate extending in a direction approaching the rotor of the motor from the first shielding plate.

2. The vehicle drive system of claim 1, wherein the first shielding plate is disposed so that a side surface of the first shielding plate opposes the gap, wherein a radially outer edge part of the first shielding plate reaches radially outward of the gap, and wherein a radially inner edge part of the first shielding plate reaches radially inward of the gap.

3. The vehicle drive system of claim 2, wherein the second shielding plate extends in a direction approaching the rotor of the motor from the radially inner edge part of the first shielding plate with respect to the shaft.

4. The vehicle drive system of claim 1, wherein the second shielding plate forms an annular shape that continues in a circumferential direction of the rotor of the motor.

5. The vehicle drive system of claim 1, wherein the second shielding plate is disposed radially inward of the gap.

6. The vehicle drive system of claim 1, further comprising a case configured to accommodate the motor and the resolver, wherein the first shielding plate is integrally formed with the second shielding plate, and wherein the first shielding plate is fastened to the case by a common fastening member with the resolver stator.

7. The vehicle drive system of claim 2, wherein the second shielding plate forms an annular shape that continues in a circumferential direction of the rotor of the motor.

8. The vehicle drive system of claim 2, wherein the second shielding plate is disposed radially inward of the gap.

9. The vehicle drive system of claim 2, further comprising a case configured to accommodate the motor and the resolver, wherein the first shielding plate is integrally formed with the second shielding plate, and wherein the first shielding plate is fastened to the case by a common fastening member with the resolver stator.

10. The vehicle drive system of claim 3, wherein the second shielding plate forms an annular shape that continues in a circumferential direction of the rotor of the motor.

11. The vehicle drive system of claim 3, wherein the second shielding plate is disposed radially inward of the gap.

12. The vehicle drive system of claim 3, further comprising a case configured to accommodate the motor and the resolver, wherein the first shielding plate is integrally formed with the second shielding plate, and wherein the first shielding plate is fastened to the case by a common fastening member with the resolver stator.